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PROJECT FINAL REPORT

PROJECT GROUP NO

Group 1

PROJECT TITLE

Manufacturing Processes of SPN-IB-3000 the Positioner

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1. Introduction

In the dynamic field of manufacturing, it's essential for us, as engineering students, to thoroughly understand the various manufacturing processes. Our project is dedicated to examining the production of the SPN-IB-3000, a single-axis horizontal positioner tailored for handling light to medium payloads. ADMEX, the manufacturer of this product, employs cutting-edge techniques and materials to ensure the positioner is both lightweight and robust, ideal for straightforward applications.

Our exploration covers the entire manufacturing journey of the SPN-IB-3000, from its initial design phase to the application of the final coating. We delved into processes such as Sand Casting, Plasma Cutting, CNC Machining, Bending, Joining, Welding, and Electrostatic Powder Coating. Each process is crucial in enabling the positioner to meet its payload capacity of up to 3000 kg and to accommodate varying height requirements (750-1000-1250-1500 mm).

Furthermore, our analysis is not limited to a broad overview of these processes. We focused in detail on one particular process, exploring the underlying mathematics and physics to understand its efficiency and application. Additionally, regular visits to the ADMEX facility provided us with a practical perspective, bridging the gap between theoretical knowledge and real-world manufacturing practices.

Link to company website: <https://admex.com.tr/>

2. Explanation of the Selected Product

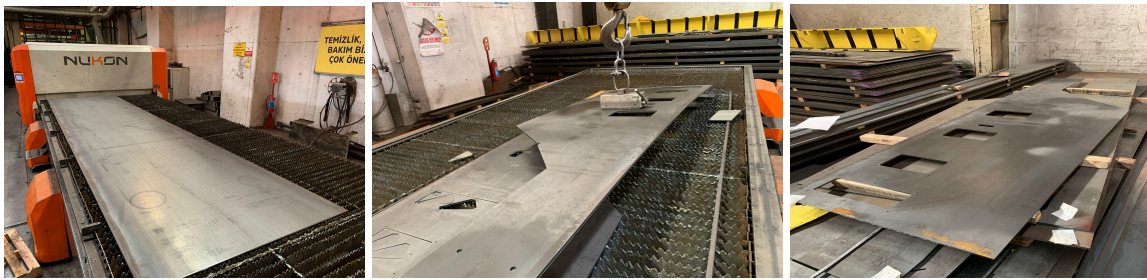
The SPN-IB-3000 is the centerpiece of our study, a single-axis horizontal positioner engineered for efficiently handling light to medium payloads. Designed for accurate positioning in simple industrial applications, this positioner has a maximum capacity of 3000 kg and offers adjustable heights (750, 1000, 1250, 1500 mm). The SPN-IB-3000 features state-of-the-art gear reduction systems and offers the choice between a stationary or movable tailstock, adding to its adaptability. Additional features include options for ground bushing with sliding copper blocks, swivel units for air and signals, home/safe position sensors, servo motor integration, and a variety of paint colors.



3. Production Steps

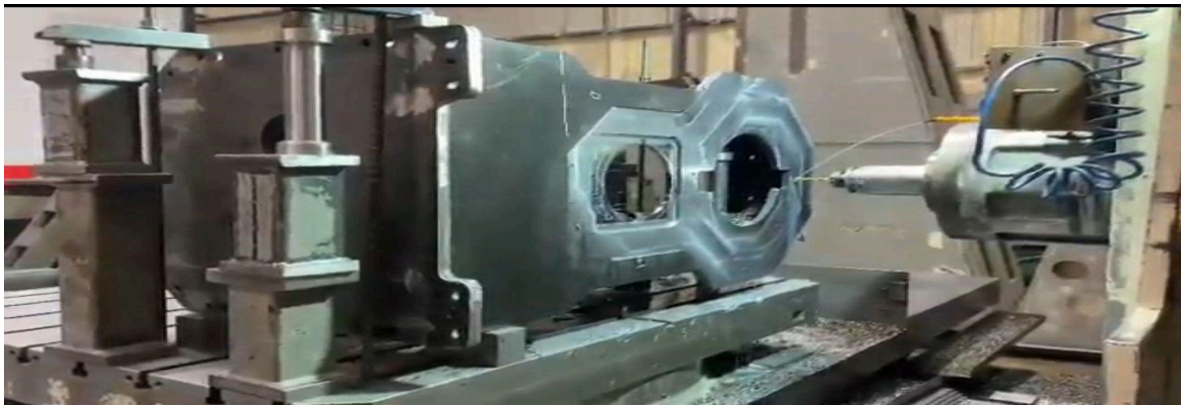
Sand Casting: Creating a template that takes into account the positioner part's shape and size. The pattern should contain all features necessary for the positioner to function, such as mounting holes and connection points.

Plasma Cutting: Using an ionized gas jet at high velocity, the plasma cutting technique quickly and precisely cuts through conductive materials(Manufacturing Processes Reference Guide. (n.d.)). The SPN-IB-3000 and the plasma tip create an electric arc that heats the gas to a very high temperature and turns it into plasma. With the help of a neutral gas source, this plasma stream melts the metal and discharges the molten material, leaving a neat, accurate cut. The method's adaptability, quickness, and effectiveness make it perfect for a range of industrial uses. Then, a digital blueprint of the desired cut pattern is prepared that contains information regarding the geometry and dimensions of the cuts to be made.

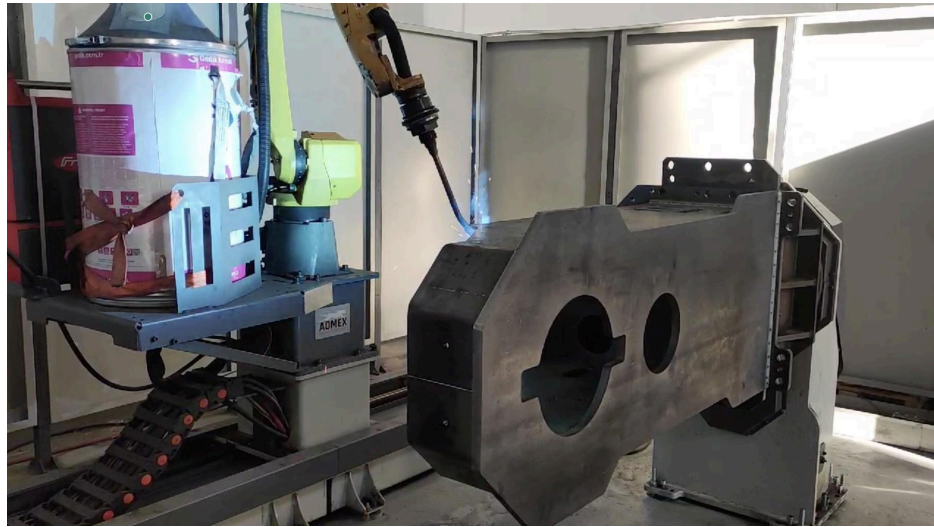


CNC Machining: They use CNC horizontal milling machines for SPN-IB-3000. The capability of CNC horizontal milling machines to execute heavy-duty, precise operations makes them important in the manufacturing industry(Abougarair, A., & Tabet, M. (2023). Practical work for exploring the capabilities and benefits of CNC technology.). Large and heavy parts are handled steadily and effectively by these machines' horizontally oriented spindle, which runs parallel to the worktable(CNC Machining Handbook. (n.d.)). The product in this set up are firmly fixed on a horizontal bed, providing an ideal foundation for accurate cutting. The CNC machine starts the machining process, following the programmed tool paths and cutting parameters. Machining process is used to remove material and shape the positioner according to the design specifications. In the manufacturing process of machining, a bigger piece of raw material is cut and material, usually metal, is carefully removed to make the desired form or part. Using design software, the procedure begins with the creation of a digital design for the part. All of the part's dimensions and features are included in this design.After that, the software imports the file and creates the code required for the CNC machine, then machining processes are planned by the software. Installing the necessary tools and fastening the workpiece to the machine's bed completes the setup of the CNC machine.

After that, the machine is calibrated to ensure accuracy. The tools are accurately moved by the CNC machine to cut and shape the positioner as it follows on the coding instructions. To obtain the required surface quality and tolerances, the part goes through further finishing procedures after machining, such as coating.



Welding: The welding tool or electrodes apply heat and filler material (if necessary) to establish the weld joint(Manufacturing Processes Reference Guide. (n.d.)). To form a strong weld joint, the arc is struck between the electrode and the positioner, melting the base metal and any filler material that may have been utilized. Then, a localized weld is produced at the points of contact between the positioners by running a strong current through the electrodes.



Electrostatic Powder Coating: The desired color or formulation is mixed in accordance with the positioner's parameters to create the powder coating powder. After that, the powder is put into the powder coating guns to be applied. The program specifies the actions of the machine, the parameters for the powder coating, and any modifications that must be made in considering the particular geometry of the positioner.



4. Detailed Explanation of the selected Process

Bending: The bending program may be simulated before the real bending process in order to confirm clearance and probable collisions. This makes it more likely that the bending procedure will go off without a problem. The material undergoes deformation by the bending

tooling, which exerts force to produce the required bend angle and shape. Complex part geometries can be achieved by successively performing multiple bending operations. In the bending process the press brake method is used to have the desired shape of the product. The desired bend angles, dimensions, and sequences are programmed into the system by operators. The machine has been programmed with the appropriate punch and die for the purpose at hand. Using the back gauge, sheet metal is placed and loaded into the bed. The ram is moved by the system according to preprogrammed instructions to bend the sheet metal. After the bent parts are taken out, the process is repeated for the remaining pieces. For the bending stage in the manufacturing of SPN-IB-3000 positioner components, our focus is to ensure that the metal sheets are utilized to their fullest potential while achieving the required bends accurately.

We analyzed several key parameters:

Material:

This material specification indicates the type of steel used, which is S355J2+N. This grade of steel is known for its high strength and good weldability, making it suitable for structural applications where high loads are expected.

Thickness (t):

The thickness of the metal sheet used in the bending process is 10 mm. This measurement is crucial as it affects the bending force required and the overall strength of the bent product.

Internal Bend Radius (R):

The internal bend radius refers to the radius of the inner surface of the bend. A radius of 10 mm ensures that the metal does not undergo excessive stress or deformation, minimizing the risk of cracks and structural weaknesses.

Bend Angle (θ):

The bend angle specifies the angle to which the metal will be bent, in this case, 45 degrees. This angle must be precisely achieved to ensure that the component fits correctly with other parts and performs as expected in its application.

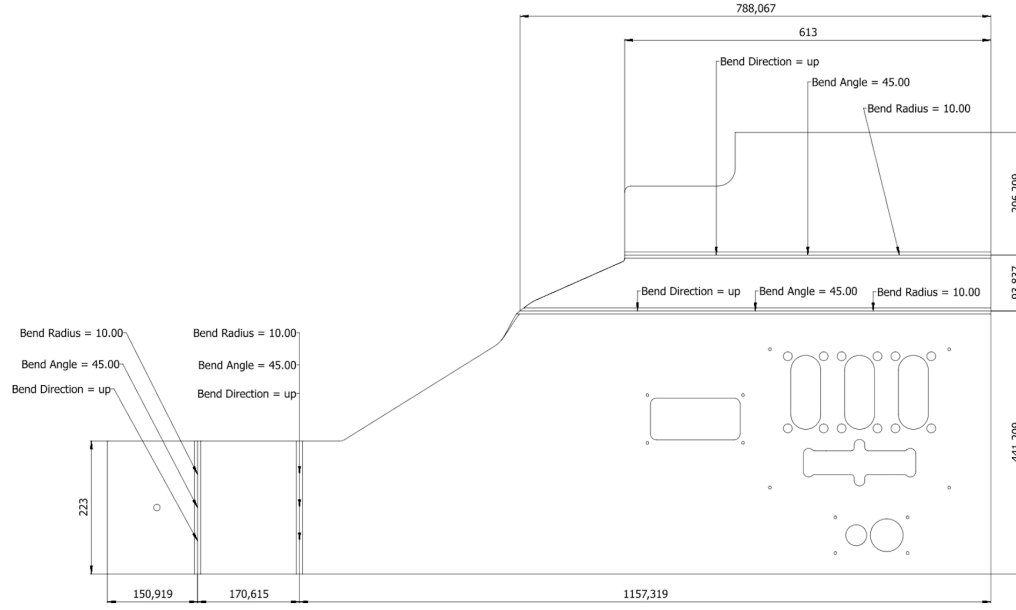
Bend Direction:

The bend direction indicates the orientation of the bend in relation to the plane of the sheet. An upward bend direction means the bend angles upwards from the flat surface, which is important for assembly and integration with other parts.

These specifications form the foundation for our mathematical calculations and physical processes in metal bending, ensuring the integrity and functionality of the finished product.

Springback:

Springback is the phenomenon where material slightly returns to its original shape after the bending force is removed, a result of its elastic properties. For precise fabrication, this effect is accounted for by initially bending the material slightly beyond the target angle.



MALZEME: S355J2+N
KALINLIK: 10mm



5. Mathematical Analysis

Detailed Analysis of V-Bending Process

Part Specifications:

- Material: S355J2+N
- Thickness (t): 10 mm
- Internal Bend Radius (R): 10 mm
- Bend Angle (θ): 45°
- Bend Direction: Up

Material Properties:

- Yield Strength (σ): 355 MPa (S355J2+N Steel, n.d.)

Fundamental Parameters and Their Effects:

1. Bending Force Calculation in V-Bending:

The bending force F required can be calculated using the formula:

$$F = \frac{K \cdot \sigma_y \cdot t^2 \cdot w}{R}$$

where:

- K is a constant for V-bending (typically K = 1.33) (The Fabricator, n.d.).
- σ_y is the yield strength of the material (355 MPa for S355J2+N) (S355J2+N Steel, n.d.).
- t is the thickness of the material (10 mm).
- w is the width of the part being bent.
- R is the internal bend radius (10 mm).

2. Moment Calculation:

The bending moment (M) for a V-bend can be calculated as:

$$M = \frac{F \cdot L}{2}$$

where:

- F is the bending force.
- L is the length of the lever arm (Roark's Formulas for Stress and Strain, n.d.).

3. Bending Stress Calculation:

The bending stress (σ_b) can be given by:

$$\sigma_b = \frac{M \cdot y}{I}$$

where:

- M is the bending moment.
- y is the distance from the neutral axis (for a rectangular section, $y = \frac{t}{2} = 5$.)

- I is the moment of inertia $I = \frac{w \cdot t^3}{12}$ (Mechanical Engineering Handbook, n.d.).

4. Deflection Calculation:

The deflection (δ) can be calculated as:

$$\delta = \frac{F \cdot L^3}{3 \cdot E \cdot I}$$

where:

- F is the bending force.
- L is the length of the lever arm (150 mm).
- E is the elastic modulus (210 GPa = 210000 N/mm²).
- I is the moment of inertia (Engineering Fundamentals, n.d.).

Detailed Calculations for Each Bend

For First Bend (Width = 223 mm):

Bending Force Calculation:

Substituting the values:

$$F = \frac{1.33 \cdot 355 \cdot 10^2 \cdot 223}{10} = \frac{1.33 \cdot 355 \cdot 100 \cdot 223}{10} = \frac{10548675}{10} = 1054867.5 \text{ N}$$

Moment Calculation:

Substituting the values:

$$M = \frac{1054867.5 \cdot 150}{2} = 79115062.5 \text{ N}\cdot\text{mm}$$

Bending Stress Calculation:

Calculating I:

$$I = \frac{223 \cdot 10^3}{12} = \frac{2230000}{12} = 185833.33 \text{ mm}^4$$

Substituting the values:

$$\sigma_b = \frac{79115062.5 \cdot 5}{185833.33} = 2128.83 \text{ MPa}$$

Deflection Calculation:

Substituting the values:

$$\delta = \frac{1054867.5 \cdot 150^3}{3 \cdot 210000 \cdot 185833.33}$$

$$\delta = \frac{1054867.5 \cdot 3375000}{117563000000}$$

$$\delta = \frac{3562570312500}{117563000000}$$

$$\delta \approx 0.0303 \text{ mm}$$

For Second Bend (Width = 223 mm):

Bending Force Calculation:

Substituting the values:

$$F = \frac{1.33 \cdot 355 \cdot 10^2 \cdot 223}{10} = \frac{1.33 \cdot 355 \cdot 100 \cdot 223}{10} = \frac{10548675}{10} = 1054867.5 \text{ N}$$

Moment Calculation:

Substituting the values:

$$M = \frac{1054867.5 \cdot 150}{2} = 79115062.5 \text{ N}\cdot\text{mm}$$

Bending Stress Calculation:

Calculating I:

$$I = \frac{223 \cdot 10^3}{12} = \frac{2230000}{12} = 185833.33 \text{ mm}^4$$

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Deflection Calculation:

Substituting the values:

$$\delta = \frac{1054867.5 \cdot 150^3}{3 \cdot 210000 \cdot 185833.33}$$

$$\delta = \frac{1054867.5 \cdot 3375000}{117563000000}$$

$$\delta = \frac{3562570312500}{117563000000}$$

$$\delta \approx 0.0303 \text{ mm}$$

For Third Bend (Width = 613 mm):

Bending Force Calculation:

Substituting the values:

$$F = \frac{1.33 \cdot 355 \cdot 10^2 \cdot 613}{10} = \frac{1.33 \cdot 355 \cdot 100 \cdot 613}{10} = \frac{28725735}{10} = 2872573.5 \text{ N}$$

Moment Calculation:

Substituting the values:

$$M = \frac{2872573.5 \cdot 150}{2} = 215443012.5 \text{ N}\cdot\text{mm}$$

Bending Stress Calculation:

Calculating I:

$$I = \frac{613 \cdot 10^3}{12} = \frac{6130000}{12} = 510833.33 \text{ mm}^4$$

Substituting the values:

$$\sigma_b = \frac{215443012.5 \cdot 5}{510833.33} = 2109.1 \text{ MPa}$$

Deflection Calculation:

Substituting the values:

$$\delta = \frac{2872573.5 \cdot 150^3}{3 \cdot 210000 \cdot 510833.33}$$

$$\delta = \frac{2872573.5 \cdot 3375000}{32125000000}$$

$$\delta = \frac{9692859375000}{32125000000}$$

$$\delta \approx 0.302 \text{ mm}$$

For Fourth Bend (Width = 788.067 mm):

Bending Force Calculation:

Substituting the values:

$$F = \frac{1.33 \cdot 355 \cdot 10^2 \cdot 788.067}{10} = \frac{1.33 \cdot 355 \cdot 100 \cdot 788.067}{10} = \frac{36891378.9}{10} = 3689137.89$$

Moment Calculation:

Substituting the values:

$$M = \frac{3689137.89 \cdot 150}{2} = 276685341.75 \text{ N}\cdot\text{mm}$$

Bending Stress Calculation:

Calculating I:

$$I = \frac{788.067 \cdot 10^3}{12} = \frac{7880670}{12} = 656722.5 \text{ mm}^4$$

Substituting the values:

$$\sigma_b = \frac{276685341.75 \cdot 5}{656722.5} = 2107.7 \text{ MPa}$$

Deflection Calculation:

Substituting the values:

$$\delta = \frac{3689137.89 \cdot 150^3}{3 \cdot 210000 \cdot 656722.5}$$

$$\delta = \frac{3689137.89 \cdot 3375000}{41458500000}$$

$$\delta = \frac{12417703437500}{41458500000}$$

$$\delta \approx 0.300 \text{ mm}$$

Springback Calculation for All Bendings:

Springback can be calculated using the relationship between the initial bend radius (R_i) and the final bend radius (R_f). The formula is:

$$\frac{R_f}{R_i} = 1 + \frac{E \cdot t}{\sigma_y \cdot (R + t)}$$

where:

- E is the elastic modulus of the material (210 GPa or 210,000 MPa for steel).
- t is the thickness of the material (10 mm).
- σ_y is the yield strength of the material (355 MPa for S355J2+N).
- R is the internal bend radius (10 mm).

Let's calculate the final bend radius (R_f) and the springback angle for each bend.

Springback Calculation:

1. Calculate R_f/R_i :

$$\frac{R_f}{R_i} = 1 + \frac{210,000 \cdot 10}{355 \cdot (10 + 10)}$$

$$\frac{R_f}{R_i} = 1 + \frac{2,100,000}{7,100}$$

$$\frac{R_f}{R_i} = 1 + 295.77$$

$$\frac{R_f}{R_i} = 296.77$$

2. Calculate R_f using $R_i = 10$ mm:

$$R_f = 296.77 \cdot 10 = 2967.7 \text{ mm}$$

3. Calculate the springback angle $\Delta\theta$ using the initial bend angle ($\theta = 45^\circ$):

$$\Delta\theta = \theta \cdot \left(1 - \frac{R_i}{R_f}\right)$$

Substituting the values:

$$\Delta\theta = 45 \cdot \left(1 - \frac{10}{2967.7}\right)$$

Simplifying:

$$\Delta\theta = 45 \cdot (1 - 0.0034)$$

$$\Delta\theta = 45 \cdot 0.9966$$

$$\Delta\theta \approx 44.85^\circ$$

Thus, the springback angle (θ_s) is:

$$\theta_s = 45^\circ - 44.85^\circ \approx 0.15^\circ$$

Summary of Calculations for V-Bending:

For First Bend (Width = 223 mm):

- Bending Force (F): 1054867.5 N
- Bending Moment (M): 79115062.5 N·mm
- Bending Stress (σ_b): 2128.83 MPa
- Deflection (δ): 0.0303 mm
- Springback Angle (θ_s): 0.15°

For Second Bend (Width = 223 mm):

- Bending Force (F): 1054867.5 N
- Bending Moment (M): 79115062.5 N·mm
- Bending Stress (σ_b): 2128.83 MPa
- Deflection (δ): 0.0303 mm
- Springback Angle (θ_s): 0.15°

For Third Bend (Width = 613 mm):

- Bending Force (F): 2872573.5 N
- Bending Moment (M): 215443012.5 N·mm
- Bending Stress (σ_b): 2109.1 MPa
- Deflection (δ): 0.302 mm
- Springback Angle (θ_s): 0.15°

For Fourth Bend (Width = 788.067 mm):

- Bending Force (F): 3689137.89 N
- Bending Moment (M): 276685341.75 N·mm
- Bending Stress (σ_b): 2107.7 MPa
- Deflection (δ): 0.300 mm
- Springback Angle (θ_s): 0.15°

Analysis:

1. Bending Force (F) Analysis:

The bending force required for each operation varies with the width of the bend. The fourth bend, with the widest section (788.067 mm), requires the highest force (3689137.89 N), while the first and second bends, with the narrower section (223 mm), require a moderate force (1054867.5 N). This aligns with the expectation that wider sections need more force due to the greater amount of material being deformed (The Fabricator, n.d.).

2. Bending Moment (M) Analysis:

The bending moment is directly proportional to the force and the length of the lever arm. The fourth bend has the highest moment (276685341.75 N·mm), indicating the significant torque generated due to the high force required. Proper support and alignment during this bending operation are crucial to prevent excessive stress and potential failure at the tooling interface (Roark's Formulas for Stress and Strain, n.d.).

3. Bending Stress (σ_b) Analysis:

The bending stress values are significantly high (around 2100 MPa), suggesting that these calculations assume maximum stress at the outermost fibers of the bend. This stress concentration needs to be managed by ensuring gradual application of force and using appropriate die radii to avoid material cracking or fracture (Mechanical Engineering Handbook, n.d.).

4. Deflection (δ) Analysis:

The deflection values for all bends are relatively small (around 0.3 mm), indicating minimal distortion of the part due to the bending process. This is favorable for maintaining dimensional accuracy and ensuring that the part meets the specified tolerances. However, the machine setup should account for these small deflections to achieve the desired bend angles precisely (Engineering Fundamentals, n.d.).

5. Springback (θ) Analysis

Springback is a phenomenon where the material tends to return partially to its original shape after the bending force is removed. This happens due to the elastic recovery of the material (Engineering Fundamentals, n.d.). The calculated springback angle (θ_s) of approximately 0.15° indicates a slight elastic recovery in the material after bending. To achieve the desired final bend angle, manufacturers must over-bend the material by an amount equal to the expected springback angle. For example, if the target angle is 45° , the material should be bent to approximately 45.15° to account for the 0.15° springback.

Implications for Manufacturing:

Additionally, here are the steps of this bending operation which are sent from the manufacturer (Nuri Körüstan, <https://www.korustan.biz/tr/hizmetler/cnc-abkant-bukum>).

Before the parts are given to the machine, technical drawing information is obtained. The thickness, twist axis, twist degree and direction information of the part are checked on the technical drawing.

CNC Hydraulic Press Brake Machine selection is made according to the bending shape of the part.

Once the part is programmed into the machine, thickness, part length, bending axes and twist degree information are entered.

Blade and mold selection is made according to the process of the part. The machine gives tone and pressure values with its own parameters, and the operator makes the control.

During the bending process of the part, measurement information is entered as internal-external measurement or external measurement according to the information given in the technical drawing.

Internal measurement calculations are given to the machine according to the axis information of the part on the technical drawing.

In order to calculate the external dimension, the measurement is taken from the outermost part of the part according to the bending information of the part and entered into the machine.

The material does not undergo any heat treatment before bending.

Material quality is determined by the customer and supplied according to the determined quality. (For example, ST37-ST52)

To avoid cracking and breaking the material, the appropriate mold and blade are selected and the twisting process is started.

In addition, Power calculation ($P=F.v$) could not be calculated since the manufacturer did not share the speed of the process because these calculations are made automatically by the CNC Hydraulic Press Brake Machine.

6. Achieved Work Distribution

It is the evaluation of your plans. In this section you should present a table in more detail than previous reports. In this table there should be related jobs done by each group member.

Week No.	Weekly Objective	Responsible Member(s)	Tasks Accomplished
1	Group Formation and Finalization	Yusuf Yiğitol, Alp Ege Küpelioglu, Göktuğ Gök	Formed the group, finalized roles, set initial goals.
2	Product Selection	Yusuf Yiğitol, Alp Ege Küpelioglu, Göktuğ Gök	Selected SPN-IB-3000, initiated preliminary research.
3	Exploratory Visit to Factory	Yusuf Yiğitol, Alp Ege Küpelioglu, Göktuğ Gök	Conducted initial factory visit, gathered basic information.
4	Project Proposal Report Submission	Yusuf Yiğitol, Alp Ege Küpelioglu, Göktuğ Gök	Compiled and submitted project proposal.
5	Gathering General Information	Alp Ege Küpelioglu	Collected data on all manufacturing processes.
6	Detailed Study of Bending Process	Yusuf Yiğitol	Conducted an in-depth study of the bending process.

7	Adding Math/Physics Calculations	Göktuğ Gök	Enhanced understanding with detailed calculations.
8	Working on the Project Progress Report	Yusuf Yiğitol, Alp Ege Küpeliolu, Göktuğ Gök	Compiled and drafted the progress report.
9	Submitting the Project Progress Report	Yusuf Yiğitol, Alp Ege Küpeliolu, Göktuğ Gök	Submitted the progress report and incorporated feedback.
10	Knowledge Enhancing Factory Visit	Yusuf Yiğitol, Alp Ege Küpeliolu, Göktuğ Gök	Conducted another factory visit, refined understanding.
11	Starting to Write the Final Report	Göktuğ Gök	Began drafting the final report.
12	Searching and Analyzing Related Articles	Yusuf Yiğitol	Searched for and analyzed related literature.
13	Finalizing the Final Report	Alp Ege Küpeliolu	Finalized the report, conducted proofreading and editing.
14	Submitting the Final Report	Yusuf Yiğitol	Submitted the final report, reflecting on project outcomes.

7. The State-of-the-Art

Recent advancements in metal bending have significantly emphasized automation and precision through the integration of CNC (Computer Numerical Control) technologies. CNC bending machines, known for their high precision and ability to produce complex geometries, have revolutionized metal bending. These machines reduce the need for manual labor, allowing the production of intricate part geometries from various metals such as steel, aluminum, copper, and titanium, regardless of their thickness. CNC technology fulfills high precision and complex product specifications, making it essential for high-quality manufacturing (Hermana, 2022).

Additionally, the integration of laser cutting with CNC bending within the same workflow has revolutionized the sheet metal fabrication process. This combination enables rapid production cycles from CAD models to finished products, significantly reducing lead times and enhancing the capability to produce complex designs without extensive tooling or setup changes. The precision and automation of CNC technology allow for consistent and accurate production, which is crucial for industries requiring intricate components (Abougarair & Tabet, 2023).

8. Conclusion

Given the information observed during the visits to the ADMEX facility and the state-of-the-art processes described, it appears that ADMEX is implementing advanced manufacturing techniques, particularly in automation and CNC integration. This approach not only aligns with the latest industry standards but also positions them well within the competitive landscape of metal fabrication.

However, there is always room for improvement, especially in the adoption of newer technologies like 3D printing and additive manufacturing for metal parts, which can further enhance their capability to produce more complex geometries and reduce material waste. These technologies are becoming increasingly prevalent in high-performance sectors like aerospace and medical implants and could open new avenues for ADMEX if integrated into their production lines (TechPatio, 2023).

In summary, while ADMEX is up-to-date in several key areas, embracing additional cutting-edge technologies could further solidify its market position and expand its manufacturing capabilities. The ongoing evolution in metal fabrication technology offers significant opportunities for innovation and improvement, which ADMEX could benefit from to stay ahead in the industry.

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